

1. Administrative & Study Identification

Full Study Title: A Phase III, Single-Arm Multicenter Study of Atezolizumab (Anti-PD-L1 Antibody) in High PD-L1 Expression, Chemotherapy-Naïve Patients With Stage IV Non-Squamous or Squamous Non-Small Cell Lung Cancer **Short Title/Acronym:** A Study of Atezolizumab in High PD-L1 Expression, Chemotherapy-Naïve Patients With Stage IV Non-Squamous or Squamous Non-Small Cell Lung Cancer / ML42606 **Protocol Number:** ML42606 **NCT Number:** NCT05047250 **Posting ID:** 35454881208951339 **Sponsor/Company Name:** Roche **Investigational Product:** Atezolizumab (Trade Name: Tecentriq) / Anti-PD-L1 Antibody **ATC Code:** L01FF05 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/tecentriq-epar-product-information_en.pdf **Phase of Development:** Phase III **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of atezolizumab in high PD-L1 expression, chemotherapy-naïve patients with stage IV non-squamous or squamous non-small cell lung cancer (NSCLC) without a sensitizing EGFR mutation or ALK translocation. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Overall Survival (OS) rate at 1-year, and OS rate at 2-year.

2.2 Background & Rationale To address the medical necessity of providing effective targeted immunotherapy for patients with advanced (Stage IV) NSCLC who have high PD-L1 expression but no targetable EGFR or ALK mutations. Atezolizumab as a monotherapy offers a chemotherapy-free first-line treatment option for this specific patient population, potentially delaying disease progression and extending survival.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** Active, Not Recruiting **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 60 participants
Number of Sites: Multicenter **Estimated Study Start:** 14-Jun-2022
Estimated Primary Completion: 30-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. ECOG performance status of 0 or 1. 2. Histologically or cytologically confirmed, Stage IV non-squamous or squamous NSCLC. 3. No prior treatment for Stage IV non-squamous or squamous NSCLC. 4. Patients who have received prior neo-adjuvant, adjuvant chemotherapy, radiotherapy, or chemo-radiotherapy with curative intent for non-metastatic disease must have experienced a treatment-free interval of at least 6 months from enrollment since the last chemotherapy, radiotherapy, or chemo-radiotherapy cycle. 5. Tumor TC3 or IC3, as determined by SP142 performed by a central laboratory on previously obtained archival tumor tissue or tissue obtained from a biopsy at screening. 6. Measurable disease, as defined by RECIST v1.1. 7. Adequate hematologic and end-organ function. 8. Life expectancy ≥ 3 months. 9. For women of childbearing potential: agreement to remain abstinent or use contraception, and agreement to refrain from donating eggs.

4.2 Exclusion Criteria 1. Known sensitizing mutation in the EGFR gene or ALK fusion oncogene. 2. Symptomatic, untreated, or actively progressing CNS metastases. 3. Spinal cord compression not definitively treated with surgery and/or radiation, or previously diagnosed and treated spinal cord compression without evidence that disease has been clinically stable for ≥ 2 weeks prior to enrollment. 4. Current leptomeningeal disease. 5. Uncontrolled tumor-related pain. 6. Uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures. 7. Uncontrolled or symptomatic hypercalcemia. 8. Malignancies other than NSCLC within 5 years prior to enrollment, with the exception of those with a negligible risk of metastasis or death treated with expected curative outcome. 9. Pregnancy or breastfeeding, or intention of becoming pregnant during study treatment or within at least 5 months after the last dose of atezolizumab. 10. History of severe allergic, anaphylactic, or other hypersensitivity reactions to chimeric or humanized antibodies or fusion proteins. 11. Known allergy or hypersensitivity to biopharmaceuticals produced in Chinese hamster ovary (CHO) cells or any component of the atezolizumab formulation. 12. Active or history of autoimmune disease or immune deficiency. 13. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, idiopathic pneumonitis, or evidence of active pneumonitis on screening chest CT scan. 14. Positive human immunodeficiency virus (HIV) test result at screening. 15. Patients with active hepatitis B or active hepatitis C at screening. 16. Active tuberculosis. 17. Severe infections within 4 weeks prior to randomization, including, but not limited to, hospitalization for complications of infection, bacteremia, or severe pneumonia. 18. Significant cardiovascular disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atezolizumab 1200 mg administered on Day 1 of each 21-day cycle. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until unacceptable toxicity or loss of clinical benefit as determined by investigators.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care and symptom management medications. **Prohibited:** Concurrent therapies targeting EGFR/ALK, other investigational agents, or systemic chemotherapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, PD-L1 testing, EGFR/ALK mutation status
Baseline	Day 1 (Cycle 1)	First Dose of Atezolizumab (1200 mg IV)
Interim	Every 21 Days	IV Infusion, Safety Labs, Efficacy Assessment (RECIST v1.1)
End of Study	Varies	Treatment discontinuation due to unacceptable toxicity or loss of clinical benefit, Follow-up for Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall survival (OS), defined as the time from atezolizumab initiation to death from any cause.

7.2 Secondary & Exploratory Endpoints 1. Progression-free survival (PFS). 2. OS Rate at 1-Year. 3. OS Rate at 2-Year.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite clinical evaluations and laboratory assessments every 21 days. **Serious Adverse Events (SAEs):** Reporting timeline, typically within 24 hours to the sponsor. **Data Monitoring Committee (DMC):** Internal/External safety monitoring mechanisms managed by Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for all enrolled patients receiving at least one dose of atezolizumab. **Sample Size Justification:** \$N=60\$ is estimated to provide adequate statistical power to evaluate the efficacy in this specific high PD-L1 expression cohort.

Handling of Missing Data: Censoring implemented at the last known alive/progression-free date for OS and PFS calculations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Frequent onsite and remote monitoring visits as per sponsor guidelines. **Audit Trail:** Details on eCRF locking and data clarification forms managed by Roche data management.

11. Study Identifiers & Governance

Primary ID: ML42606 **Registry Number:** NCT05047250 **Posting ID:** 35454881208951339 **Sponsor/Lead Organization:** Roche **Study Title:** A Study of Atezolizumab in High PD-L1 Expression, Chemotherapy-Naïve Patients With Stage IV Non-Squamous or Squamous Non-Small Cell Lung Cancer **Official Regulatory Title:** A Phase III, Single-Arm Multicenter Study of Atezolizumab (Anti-PD-L1 Antibody) in High PD-L1 Expression, Chemotherapy-Naïve Patients With Stage IV Non-Squamous or Squamous Non-Small Cell Lung Cancer

12. Clinical Framework (NSCLC Specific)

Primary Condition: Metastatic Non-Small Cell Lung Cancer (Neoplastic Process) **Preferred UMLS Name:** Atezolizumab for Non-Small Cell Lung Carcinoma **Diagnosis Threshold:** High PD-L1 expression, Chemotherapy-naïve, without EGFR/ALK mutations (Stage IV) **Medical Methodology:** Atezolizumab Immunotherapy Monotherapy **Optimization Target:** Disease progression delay and overall survival extension

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Stage IV NSCLC immunotherapy **Education Protocol (HCP):** Protocol-specific training for Atezolizumab administration and toxicity management **Education Protocol (Patient):** Informed consent, treatment compliance, and AE self-reporting instructions **Quality Audit Mechanism:** Source data verification and continuous clinical monitoring

14. Observation & Follow-Up Schedule

Total Duration: Until loss of clinical benefit or unacceptable toxicity (Estimated through late 2027) **Onsite Visit Cadence:** Every 21 days (Cycle Day 1) **Remote Monitoring Frequency:** Standard

protocol-defined monitoring **Checklist Review Interval:** Every 21 days prior to infusion

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					